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Themes and Projects Overview

Human Computer Interaction

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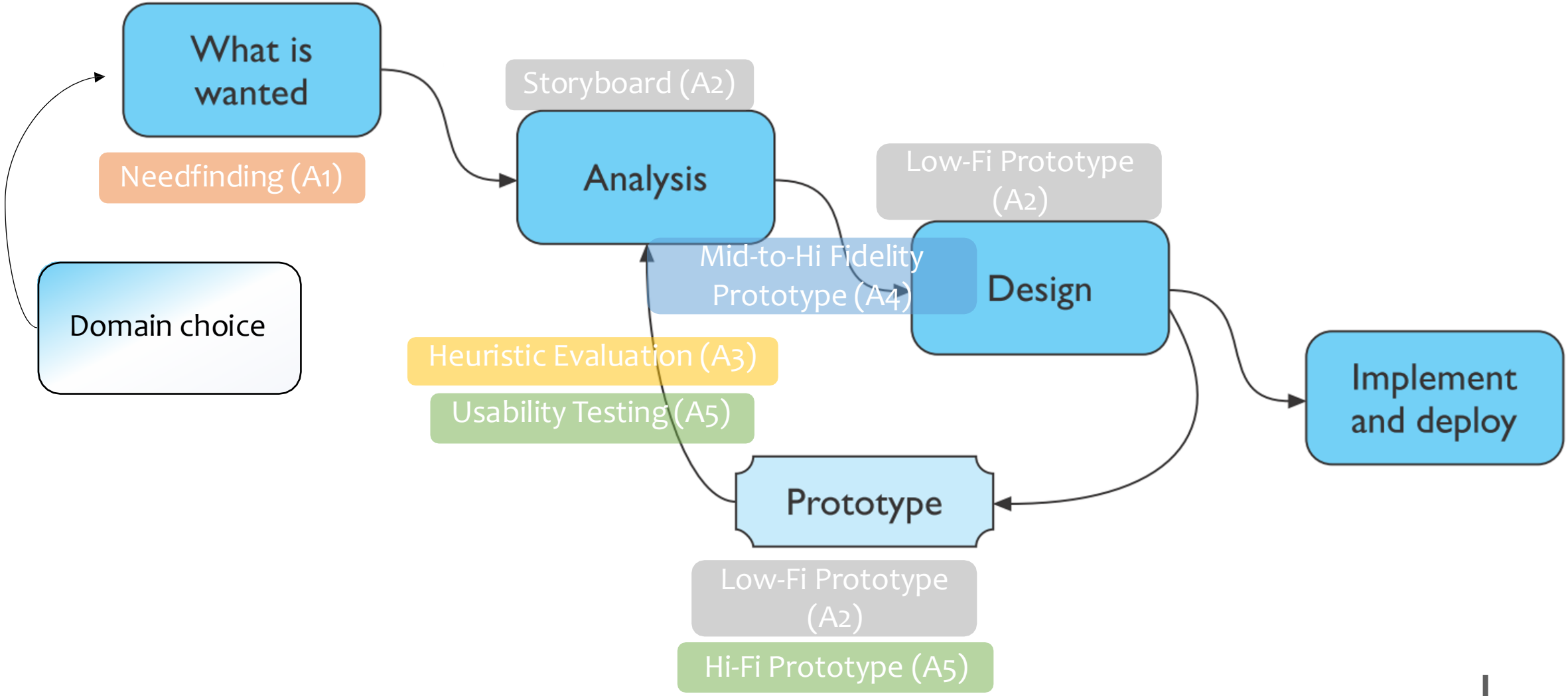
Project Development – Recap

- **Goal:**
  - to give hands-on experience with the modern human-centered design process described during the course
- Projects will be built **step-by-step** and mostly carried on *during* labs
- Project's topic proposed by each group
  - Within the chosen theme
  - Based on *needfinding*
- *Group assignments* represent the various process steps
  - Start during a lab
  - Are often followed by *checks* with teachers (in one of the following labs)
  - Evaluated at the exam through reports and discussion

(Planned) Assignments – Recap

- Assignment 1 [group]
  - *Needfinding*
- Assignment 2 [group]
  - *Storyboard and Low-fidelity prototype*
- Assignment 3 [individual]
  - *Heuristic evaluation*
- Assignment 4 [group]
  - *Medium-fidelity prototype*
- Assignment 5 [group]
  - *High-fidelity prototype and usability evaluation*

Human-Centered Process vs. Assignments



Projects Completion Level (I)

- The realized final prototype *must* be a **high-fidelity interactive prototype**
  - in code, with any technology you like/know
  - not a final "product"
  - simulating a realistic experience on a specific device

- For example, you can use web technology to built a prototype that *looks like* and *behave as* a mobile application

Projects Completion Level (II)

- The prototype will make some assumptions and won't implement standard or very complex features (even if technically important)
  - Which will be *faked*, *limited* in number, or *hard coded*
  - Your users will be already registered, signed in, and ready to "work" (no
  - sign up, sign in, … mechanism)
  - Any particular behavior will work in a small number of pre-defined cases
  - …

This Year's Themes

Projects (on laboratories)

- Themes (vertical themes):
  1. **Health and Wellbeing**
  2. **Environmental sustainability**
  3. **Smart Cities & Urban Life**

Theme 1 – **Health and Wellbeing**



Description: Health and wellbeing are fundamental aspects of our daily existence. They encompass physical, mental, and emotional dimensions that can be significantly influenced by our interactions with technology. Within this theme, you will explore innovative ways to create applications and interfaces that **empower individuals to lead healthier lives**.

Theme 1: Perspectives

- In thinking at the theme, you can ask yourselves:
  - Which health or wellbeing related topics we want to explore?
  - For whom?
  - Are we able to reach out to these people?
- Creativity and originality are *criteria* for the evaluation!
  - Helping students attending a local gym is less original/creative than supporting them in using their smartphones mindfully and effectively managing their digital habits

Theme 1 – **Health and Wellbeing**

- Examples:
  - Digital wellbeing and healthy technology use
  - Mental wellbeing and stress management
  - Physical activity, sport and fitness
  - Sleep and healthy routines
  - Personal health tracking and self-monitoring
  - Healthy Nutritions
  - …..

Theme 2 – **Environmental sustainability**



Description: Environmental sustainability refers to using resources in ways that meet present needs without compromising the ability of future generations to meet theirs. It involves protecting ecosystems, reducing pollution, protecting biodiversity, conserving energy and water and shifting toward practices that minimize harm to the planet. Within this theme, we will explore innovative ways to create systems and technology for a **more sustainable future, while increasing people awareness**!

Theme 2: Perspectives

- In thinking at the theme, you can ask yourselves:
  - What can be done for increasing environmental sustainability?
  - Are people aware about environmental sustainability?
  - What can be done to make people more aware?

- Creativity and originality are *criteria* for the evaluation!

Theme 2 – **Environmental sustainability**

- A digital twin application for water monitoring
- A IoT infrastructure for environmental monitoring
- A camera-bases system for fauna detections
- A DIY 3D-printed bees house app for biodiversity monitoring and awareness
- An AI-powered infographic for environmental awareness
- A mobile app for citizen science
- ….
- …

Theme 3 – **Smart Cities & Urban Life**



Description: Cities are complex environments where people, technology, infrastructure, and public services constantly interact. Digital technologies can play an important role in improving how we move, communicate, access services, and experience public spaces. Within this theme, you will explore innovative ways to create applications and interfaces that make urban life more accessible, connected, sustainable, and people-centered.

Theme 3: Perspectives

- **In thinking about the theme, you can ask yourselves:**
  - What challenges do people face in their everyday urban life?
  - How can technology make cities more accessible, inclusive, and livable?
  - How can we improve the way people move around and navigate the city?
  - How can technology help people discover and access urban services and public spaces?
  - How can citizens actively contribute to improving their city?
  - …

Theme 3– **Smart cities & Urban life**

- Accessible wayfinding for people with disabilities
- Sustainable smart mobility
- Personalized wayfinding for runners
- Finding accessible and inclusive public spaces
- Connecting citizens with local services
- Supporting citizen reporting of urban issues
- Discovering less crowded routes and places
- …
- …

Projects (on laboratories)

- Themes (horizontal themes):
  1. Playfull interaction or gamification
  2. Human-AI interaction

You must include at least one of these horizontal themes when implementing your solution! (both are also fine, but please remember to be sure users will need it/them!!)

Games (in brief)

- Three central elements ~~to play and~~ about games:
  1. **Dynamics**. General elements that define the patterns of how the game and the players will evolve over some time. The most popular game dynamics are narrative (good and engaging storyline), progression, and relationship.
  2. **Mechanics**. Action that moves the game forward, things that "makes the game an exciting and engaging one" like challenges, discoveries, collecting items, and rewards.
  3. **Game components**. Specific features that represent the intended mechanics and dynamics, like points, badges, and leaderboard.

We go on this about gamification

We focus only on basic elements like leaderboard, badges etc etc (gamification)

MDA framework

used to create games

- In game design the **Mechanics-Dynamics-Aesthetics** (MDA) framework is a tool used to analyze games.



- Mechanics - base components of the game, rules, algorithms and data structures in the game engine etc.
- Dynamics – interaction mechanics and players.
- Aesthetics - emotional responses of players
- Mechanics of MDA framework is very important in game design.

https://users.cs.northwestern.edu/~hunicke/MDA.pdf

Games and Gamification (in brief)

- Three central elements to play and games:
  1. **Dynamics**. General elements that define the patterns of how the game and the players will evolve over some time. The most popular game dynamics are narrative (good and engaging storyline), progression, and relationship.
  2. **Mechanics**. Action that moves the game forward, things that "makes the game an exciting and engaging one" like challenges, discoveries, collecting items, and rewards.
  3. **Game components**. Specific features that represent the intended mechanics and dynamics, like points, badges, and leaderboard.
- *Gamification*: the application of certain game mechanics to provide extrinsic motivation by offering intangible rewards, without other dynamics, to create user engagement.

Gamification

Framework about Gamification

All the things that you can have in a gamified system (not all)



-Status Points
-Badges (Achievement Symbols)
-Fixed Action Rewards (Earned Lunch)
-Leaderboard
-Progress Bar
-Quest Lists
-Dessert Oasis
-High Five
-Crowning
-Anticipation Parade
-Aura Effect
-Step-by-Step Overlay Tutorial
-Boss Fights

-Narrative
-Elitism
-Humanity Hero
-Revealed Heart

-Beginners Luck
-Free Lunch
-Visual Storytelling
-Co-creationist

-Milestone Unlocks
-Real-Time Control
-Chain Combos
-Dynamic Feedback
-Boosters
-Blank Fills
-Plant Pickers
-Poison Pickers
-Attribute Web Chart
-Double Edged Sword

-Exchangeable Points
-Virtual Goods
-Build from Scratch
-Alfred Effect
-Collection Sets
-Avatar
-Protector Quest
-Pet Companion
-Observer Attachment

-Friending
-Social Treasure/Gifting
-SeeSaw Bump
-Group Quests
-Trophy Shelf
-Brag Button
-Water Cooler
-Conformity Anchors
-Mentorship
-Social Prod

-Appointment Dynamics
-Magnetic Caps
-Dangling
-Prize Pacing
-Bootleg Quest
-Last Mile Drive
-Count Down Timer
-Torture Breaks
-Moats
-The Big Burn

-Sunk Cost Prison
-Progress Loss
-Rightful Heritage
-Evanescence Opportunity

-Status Quo Sloth
-Streaking
-Evil Egg
-FOMO Punch

-Glowing Choice
-MiniQuests
-Visual Storytelling
-Easter Eggs
-Random Rewards
-Obvious Wonder
-Rolling Rewards
-Evolved UI
-Sudden Rewards
-Oracle Effect

What About Playfulness?

- Playfulness: an attitude of mind
  - according to John Dewey, a philosopher
- *Gamification and games are manifestations of playfulness, but they not always create a playful mindset or experience on their own*
- Playfulness is more about *dynamics*, the spontaneous, creative experience, and less on mechanics (which can be still present)
  - E.g., it can be found in humor, roaming around in a virtual environment or periodic reflection challenges
  - It increase effectiveness, motivation, and creativity
- In this theme, we want our users to manifest this mindset as much as possible

Human-AI Interaction vs Human-Computer Interaction?

they are different

- What is different when AI is in an interactive system?
- AI-based systems are typically performed under **uncertainty**
  - often producing false positives and false negatives
- Many AI components are inherently **inconsistent**
  - they may respond differently to the same text input over time (e.g., autocompletion systems suggesting different words after language model updates)
  - or behave differently from one user to the next (e.g., search engines returning different results due to personalization)



How Can We Design Interactivity with AI?

can be faked

Study the interaction of user with AI

- In brief, case by case, feature by feature:
  - Decide when "to AI" and when "not to AI"
  - Understand when to automate (i.e., replace the user) and when to augment users' capabilities
  - Balance the uncertainty of AI systems with proper expectations and feedback
- In general, the more **critical** a functionality is, the more people *need* accurate and reliable results
- On the other hand, if a **complementary** feature delivers results that are not always of the highest quality, people *may* be more forgiving



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- These slides are a modified version of the ones of prof. Luigi De Russis, PoliTo. A big thanks to Luigi De Russis (PoliTo) for his work!